Fusion Sheet Metal Nesting

Fusion add-in — user guide, version 1.0.0

What does it do?

It gets the parts you drew as sheet metal in Fusion ready for laser cutting, in one click:

- Flattens every visible sheet metal body. **Your design is never changed**; all work happens in temporary copies.
- Writes each part as a DXF and nests them all on sheets with as little scrap as it can.
- **Sheet metal**: cut lines and red bend lines.
- **Model making** (greyboard, foam board, paper, PP, acrylic, wood): score lines where it bends, living hinges for wood, heat-bend marks for acrylic.
- Also gives a picture of the layout (PNG), a cut-only file for the laser, scrap figures and, if you want, a cost estimate.

Before you start

- The parts must be drawn as **sheet metal** in Fusion, each part its own body.
- Only **visible** bodies are taken. Hide the ones you want left out in the Browser.
- Parts of different thicknesses are grouped automatically; each thickness is nested on its own sheet.
- Output: **Documents > Fusion_SheetMetal_DXF > <document>_<date-time>**. Each run makes a new folder; old ones are kept. The folder opens by itself when the run ends.

The dialog, top to bottom

Click the button on the **SHEET METAL** tab (also under UTILITIES > ADD-INS). Hover over any setting to see what it does. The dialog remembers your last settings.

Section	What it does
Dil / Language	Turkish or English; applies when the dialog is reopened.
Made for	Sheet metal or Model making. Settings of the other mode are hidden.
Thickness sections	One section per thickness: **Include**, **Material**, **Sheet size** (first: "Automatic", the smallest sheet that takes all parts), **Change thickness** (recomputes the flat pattern at another thickness). Model making adds **Kerf**, **Bends** and **Score spacing**. **Calculate cost** shows the price fields.
Nesting	Method (True shape: least scrap; Rectangle: fast), Layout (Least scrap / Compact square / Strip), Rotation, Quality (Fast ~30 s, Balanced ~1-3 min, Best ~10 min), small parts into holes.
Spacing and kerf	Gap between parts (default: the sheet thickness, at least 3 mm), edge margin. Sheet metal: kerf and common-line cutting; model making: merge edges and the "add half kerf" ticks.
Outputs	Which files are written, each its own tick.
Line look	Colour and line type of each kind of line; a kind turned off is left out.

Sheet metal, step by step

1. Open the design; the bodies to cut must be visible.

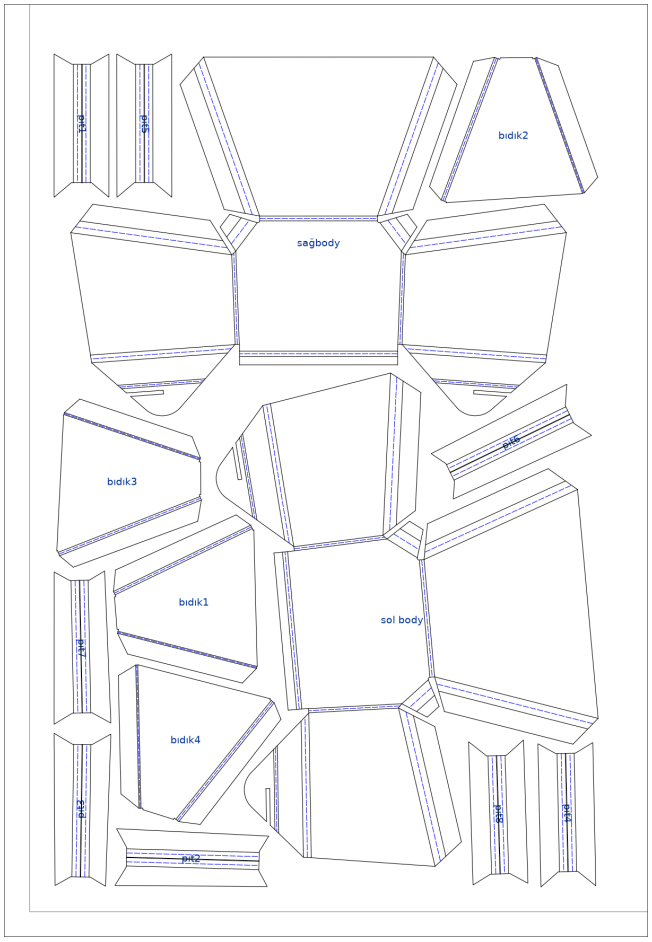
2. Click the button. **Made for** = Sheet metal.

3. In each thickness section pick the **Material** and **Sheet size**. Not sure? Leave "Automatic".

4. **Nesting**: True shape + Least scrap + Automatic rotation + Balanced suits most jobs.

5. OK. A progress window shows the time taken; "Cancel" switches to fast rectangle nesting.

6. At the end a summary appears (parts, scrap, cost if priced) and the output folder opens.

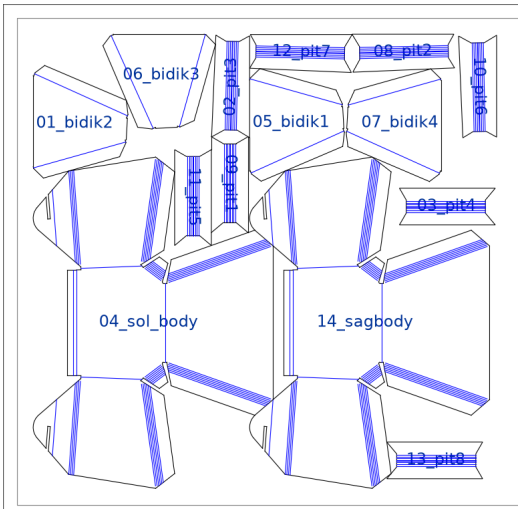


Sheet metal example: black cut lines, red bend lines; part names only in the picture (no text in the DXF, the laser would cut it).

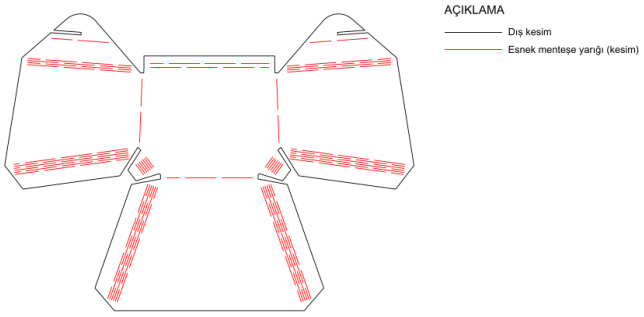
Model making, step by step

1. **Made for** = Model making.
2. Pick the **Material**: greyboard, paper/Bristol, PP, corrugated board, foam board, balsa, plywood/MDF, acrylic, Styrofoam. Kerf, bend handling and score spacing fill in to suit it.
3. **Sheet size**: shop sizes (A4, A3, 350×500, 500×700, 700×1000) and laser bed sizes; "Automatic" picks the smallest that fits.
4. Prototyping a sheet metal part? Pick the same thickness as the metal and leave "Change thickness" off: the model's flat pattern is then the metal one.
5. OK. The cut file holds the cut and score lines together.

Bends	What it draws	For
Score	Fills the bend strip with parallel score lines (round bend); a sharp bend gets one line in the middle. At least two kerfs apart, never past the part's edge.	Greyboard, paper, PP, corrugated, foam board, balsa
Living hinge	Staggered short slits (15 mm slit, 3 mm bridge); the wood flexes and never falls apart.	Plywood, MDF
Heat bend	Marks where the heated strip starts and ends; no cut.	Acrylic
No bend lines	Nothing.	Styrofoam (no laser)



Greyboard cut file: black = cut, blue = score.



Plywood: red = living-hinge slits.

Which colour gets what in the laser software?

Laser software (LightBurn, RDWorks and so on) separates lines **by colour**. Give each colour its own power and speed.

Line	Colour	On the laser
Outer cut, inner cut (hole)	Black / white	**Cut** (full power).
Living-hinge slit	Red	**Cut**.
Score	Blue	**Low power, high speed**: halfway into the material, not visible from the back. At cutting power the part falls apart at the bend. Test on scrap first.
Heat-bend mark	Green	Very low power (a light mark) or skip it.
Bend line (sheet metal)	Red	**Not cut**; for the press brake. Not in the cut file.

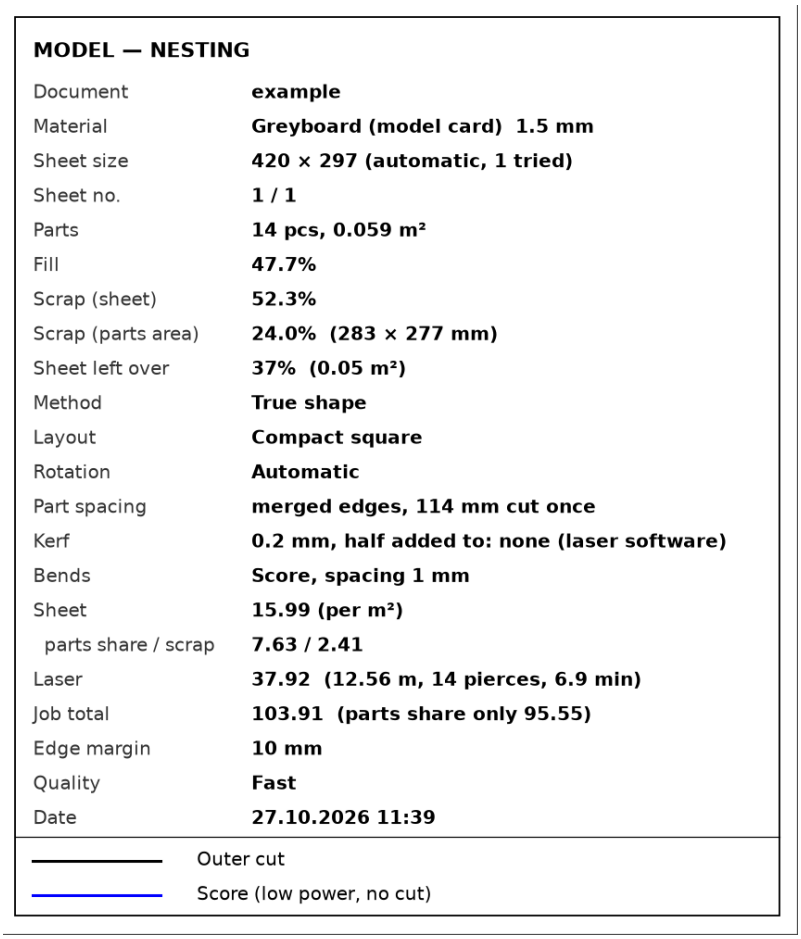
Kerf: the width the beam burns. Most laser software compensates it (LightBurn: "Kerf offset" per layer, default 0). So the add-in leaves lines at true size. If your software has no such setting, turn on "Add half kerf to outer lines / holes".

Layer names: OUTER_PROFILES / INTERIOR_PROFILES (cut), BEND / BEND_EXTENT (bend), CIZIK (score), MENTESE (living hinge), ISARET (heat-bend mark).

Output files

File	What
00_CUT_….dxf	**The file for the laser.** Cut and score lines only.
00_NESTING_….dxf	The layout with bend lines and the sheet frame.
00_NESTING_….png, 00_CUT_….png	Pictures: part names, title block at the bottom right. "_yakin": a close-up when the parts take a small corner.
00_COST_….csv	Cost part by part; opens in Excel.
01_….dxf, 02_….dxf …	Each part on its own.
_parcalar.json, _log.txt	Part list and run log. Attach _log.txt when reporting a problem.

Reading scrap and cost



Title block (greyboard, automatic A3, priced).

Row	Meaning
Fill	The parts' net area / the whole sheet.
Scrap (sheet)	Unused part of the whole sheet; high when the parts take a corner.
Scrap (parts area)	Waste inside the rectangle the parts take. **The figure to watch.**
Sheet left over	What is left outside it, reusable.
Sheet / parts share / scrap	The sheet's price; the share of the parts' area; the price of the scrap in the parts area.
Laser	Laser charge, line length, pierces, and time when priced per minute.
Job total	With whole sheets; in brackets with the parts' share only.

Entering prices

- Turn on **Calculate cost** in the thickness section. No prices, no cost.
- **Price unit**: per kg or per m² for metal; per m² for model making. Per sheet too when you pick the sheet.
- **Laser charge**: "Metres + pierces" (metal shops) or "Minutes" (model workshops; speeds are rough guesses).
- Bending (press brake) for metal and a set-up fee per job.
- Prices are remembered per material + thickness. None are filled in for you.

Troubleshooting

What happened	What to do
No button	UTILITIES > ADD-INS > Scripts and Add-Ins > Add-Ins: SacAcinimDXF, Run; tick "Run on Startup".
"No visible sheet metal bodies found"	The bodies are not sheet metal, or hidden.
"INCOMPLETE FLAT PATTERN"	Fusion could not fully flatten that part; check the part and the DXF's size.
"SKIPPED ... Flat Pattern"	Delete that component's Flat Pattern in the Browser, run again.
"THICKNESS NOT CHANGED"	Fusion could not rebuild the model at the new thickness; those parts come out at the model's thickness, in a separate file.
Nesting takes too long	Quality: Fast, or Cancel.
"No Python with ezdxf found"	Run the installer again.
Anything else	Open an issue on GitHub with a screenshot and _log.txt.

Known limits and responsibility

- Intel Macs: no true-shape nesting; parts are nested as rectangles.
- The Windows installer has not been tested on a real Windows PC yet.
- "Merge edges" only merges straight edges that lie exactly on top of each other; usually few do.
- Card does not bend like metal: a model can be off by roughly half a millimetre per bend (an estimate, not measured).
- Never laser-cut PVC / Forex (toxic chlorine gas).

This software is provided "as is" under the MIT License, without warranty of any kind. It may contain bugs. Check every DXF's dimensions before cutting; you are responsible for what you cut, the material and using your laser safely.

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